

Fabric ShoppingMart

An end-to-end retail lakehouse that brings structured and unstructured shopping data into one medallion architecture for repeatable transformation and business-ready analytics.

The challenge

Retail activity was split across order, customer, product, review, social-media, and web-log files. The project needed a unified path from raw GitHub data to clean, conformed tables and focused retail KPIs.

STRUCTURED CSV

customers.csv · products.csv · Orders_Data.csv

UNSTRUCTURED JSON

reviews.json · social_media.json · web_logs.json

The medallion design separates raw, conformed, and analytical data

Bronze

RAW

Fabric Data Factory copies source files as-is into lakehouse folders such as Bronze_Orders, Bronze_Customers, Bronze_Reviews, and Bronze_Social_Media.



Silver

CONFORMED

PySpark reads OneLake through abfss paths, removes key-column nulls, casts OrderDate, and joins orders, customers, and products into OrderDetails. Reviews are also processed.



Gold

ANALYTICS

PySpark and Spark SQL calculate business aggregations and write the results as Delta tables for downstream analysis.

1 Metadata removes pipeline rewiring

ShoppingMart_StructuredMetaData.json and ShoppingMart_UnstructuredMetaData.json map every source file to its Bronze sink folder and file. Adding another source requires a new JSON entry, not a pipeline change.

2 One master pipeline controls the complete run

Master_Pipeline_shoppingMart orchestrates four activities in sequence:

Structured ingestion

Unstructured ingestion

Silver notebook

Gold notebook

Gold tables answer practical retail questions

Product rating

Average rating for each product from review data

Social sentiment

Sentiment trends summarized by platform

Web engagement

Activity grouped by user, page, and action

Order activity

Orders per customer for each order date

What this project demonstrates

- Medallion lakehouse design
- Metadata-driven ingestion
- Fabric pipeline orchestration
- OneLake data access
- PySpark transformation
- Delta table delivery

Technology: Microsoft Fabric · OneLake · Data Factory Pipelines · PySpark · Spark SQL · Delta Lake

Built as a personal portfolio project to demonstrate hands-on Microsoft Fabric data engineering skills.